

Basic Concepts in Biomechanics

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Biomechanics is an interdisciplinary field that uses the principles of mechanics to improve the human body through design, development, and analysis of equipment, systems, and therapies. This biomechanical knowledge can help in understanding the loading of the musculoskeletal system and its mechanical responses, which can be used to determine normal function, predict changes, and propose interventions. More specifically, basic biomechanics explores forces and moments required for movement and balance of the human body by muscle recruitment and the consequence of internal forces on loading of soft tissues. This chapter explores biomechanics in terms of four topics: statics, dynamics, mechanics of materials, and applications. Additionally, examples of these concepts are provided throughout the chapter to illustrate how they can be directly applied to sports medicine.

BASIC CONCEPTS

Units of Measure

Understanding the basic units of measure for biomechanics is important in describing the dimensional or spatial analyses. There are three primary dimensions—length, time, and mass—from which secondary dimensions are subsequently derived (Table 2.1). Metric unit measurements are commonly used in the field of biomechanics to describe such dimensions and are also used in this chapter. One measurement not described in Table 2.1 is that used for angular descriptions. The unit for angles is typically defined in terms of radians or degrees.

Scalars and Vectors

Most of the physical quantities encountered in mechanics are either scalars or vectors. Scalar quantities describe only the magnitude of an element and are used for concepts such as time, length, speed, mass, temperature, volume, work, power, and energy. In contrast, both a magnitude and a direction are associated with vector quantities. Examples of vector quantities are displacement, force, moment, velocity, and acceleration. The magnitude of an object's velocity is identical to its speed, but the velocity vector also contains information about the direction of motion. Graphically, a vector is represented as an arrow, with the orientation of the arrow indicating its line of action or direction. For example, the force of gravity is a vector quantity that can be represented by a downward arrow at an object's center of mass. Typically vectors are represented in a Cartesian coordinate system.

Coordinate Systems

A coordinate system is a frame of reference for a structure or system. Coordinate systems are spatial tools that are useful in describing body position and orientation and the directionality of all vector quantities. To apply mechanical principles to a system, all vector quantities must be expressed with respect to the same coordinate system. Several types of coordinate systems are available, but the most commonly used in sports medicine research is the Cartesian coordinate system. This system can be either two- or three-dimensional. All axes are perpendicular to one another (a condition known as orthogonality) and have both positive and negative components. A vector quantity contains a component in each dimension corresponding to the magnitude associated with each axis. In a two-dimensional system, the horizontal and vertical axes are typically labeled “x” and “y,” respectively. In a three-dimensional system, the “z”-axis is added. This axis lies perpendicular to both the x- and y-axes, with its positive side determined by the right-hand rule. This rule states that if you align the index finger of your right hand with the positive x-axis and the middle finger with the positive y-axis, your thumb will be pointing in the positive direction of the z-axis. Similarly, the right-hand rule can be used to determine the positive and negative directions of angles and moments about a specified axis. While the thumb of your right hand is aligned with the rotational axis, your fingers will curl in the positive direction of rotation. These rules are very important in establishing the correct spatial quantities within a designated coordinate system.

Anatomic Coordinate System

An anatomic coordinate system is a Cartesian coordinate system with each axis representing anatomic directions such as superior-inferior, medial-lateral, and anterior-posterior. The corresponding rotations about these axes are represented by internal-external rotation, flexion-extension, and abduction-adduction, respectively. Researchers and clinicians use the anatomic coordinate system whenever possible because it is the most clinically relevant for diagnosis and treatment.

Degrees of Freedom

Degrees of freedom are the total number of independent movements needed to completely describe the position and orientation of a body in a given coordinate system. The musculoskeletal system has numerous degrees of freedom through which countless

Abstract

Biomechanics is an interdisciplinary field that uses the principles of mechanics to improve the human body through design, development, and analysis of equipment, systems, and therapies. This biomechanical knowledge can help in understanding the loading of the musculoskeletal system and its mechanical responses, which can be used to determine normal function, predict changes, and propose interventions. More specifically, basic biomechanics explores the forces and moments required for movement and balance of the human body by muscle recruitment and the consequence of internal forces on loading of soft tissues. This chapter explores biomechanics in terms of four topics: statics, dynamics, mechanics of materials, and applications. Additionally, examples of these concepts are provided throughout the chapter to illustrate how they can be directly applied to sports medicine.

Keywords

biomechanics
forces motion
material properties
viscoelasticity

TABLE 2.1 Dimensional Quantities and Corresponding Units in the International System of Units

Dimensions	SI Unit
Primary	
Length	Meter (m)
Time	Second (s)
Mass	Kilogram (kg)
Secondary	
Area	m ²
Volume	m ³
Velocity	m/s
Acceleration	m/s ²
Force	Newton (N) = kg • m/s ²
Pressure/stress	Pascal (Pa) = N/m ²
Moment/torque	N • m
Work/energy	Joule (J) = N • m
Power	J/s

SI, International System of Units.

movements are accomplished. These movements can be performed in three-dimensional space through a set of translations and rotations and are typically described in an anatomic coordinate system. For example, the glenohumeral joint has a total of six degrees of freedom: three translations (superior-inferior, medial-lateral, and anterior-posterior), and three rotations (internal-external, abduction-adduction, and flexion-extension). However, depending on the analysis, the glenohumeral joint can be assumed to be a ball-and-socket joint, thus reducing the number of degrees of freedom to three rotations by constraining the translations. Similarly, the elbow only allows for one degree of freedom when it is considered a simple hinge joint.

Degrees of freedom are important to consider when describing body motion and position numerically. Joint motion may be constrained by the shape of the articular surfaces or by stabilizing ligaments, reducing the degrees of freedom and simplifying the description. However, some joints, such as the knee and ankle, also allow for small amounts of translation that are important to consider when interpreting a clinical examination. With an anterior cruciate ligament (ACL) injury, the knee will have an increased level of anterior translation, thereby altering the anterior degree of freedom. Using reduced degrees of freedom is strategic when modeling or characterizing joints of the body. For example, when a person picks up an object from the floor and subsequently returns to an erect posture, each vertebra moves in three-dimensional space to achieve the desired posture, representing a large number of degrees of freedom. To evaluate lower back injuries using modeling techniques, the degrees of freedom can be significantly reduced by only considering the L5–S1 joint and assuming that the motion occurs only in the sagittal plane, resulting in a two-dimensional analysis. Similarly, in assessing the tibiofemoral joint during a kicking motion, it could be assumed that the tibia moves in only two dimensions within the sagittal plane (Fig. 2.1A). These reductive assumptions often overestimate the translations, rotations, forces, and moments required to perform the motion.

TABLE 2.2 Newton's Laws of Motion Applied to Static and Dynamic Analyses

Law	Definition	Equation
First (inertia)	An object at rest tends to stay at rest and an object in motion tends to stay in motion with the same speed and in the same direction unless acted upon by an external unbalanced force.	$\Sigma F = 0$ $\Sigma M = 0$
Second (acceleration)	The rate of change of the moment of a body is directly proportional to the applied force and takes place in the direction in which the force acts.	$F_{\text{NET}} = m \cdot a$ $M_{\text{NET}} = I \cdot \alpha$
Third (action-reaction)	For every action there is an equal but opposite reaction.	

Newton's Laws

Three basic rules of physics—Newton's laws of motion—are used to describe the relationship between the forces applied to the body and the consequences of those forces on human motion (Table 2.2).

Newton's First Law (Inertia)

Newton's first law states that an object at rest tends to stay at rest and an object in motion tends to stay in motion with the same speed and in the same direction (velocity) unless acted upon by an unbalanced force. This law is also commonly referred to as the *law of inertia*. Therefore a nonzero resultant force must act on a rigid body to change its velocity. For example, headrests are placed in cars to prevent whiplash injuries during rear-end collisions by stopping the motion of the head, which illustrates decreasing linear velocity and acceleration by application of an unbalanced force. Furthermore, if the resultant moments and forces are zero, then the body will also have no rotational or linear acceleration.

Newton's Second Law (Acceleration)

Newton's second law pertains to the behavior of objects for which all forces are not balanced. Therefore the resultant force is not equal to zero, and acceleration in the direction of the applied force will occur. The magnitude of the acceleration is proportional to the magnitude of the resultant forces or moments applied to the body. Thus, in essence, the first law is a special case of the second law:

$$\Sigma F = m \cdot a : \Sigma M = I \cdot \alpha$$

where “F” is the resultant force, “m” is the mass of the body being acted upon, “a” is the acceleration of the body due to the unbalanced external forces, “M” is the resultant moment, “I” is the mass moment of inertia (resistance of a body to rotation), and α is the angular acceleration.

Linear acceleration can exist as an extremity is sent through a range of motion (ROM) during a task (see Fig. 2.1F). In sports, an athlete frequently controls his or her mass moment of inertia or center of mass of the entire body by altering the positioning

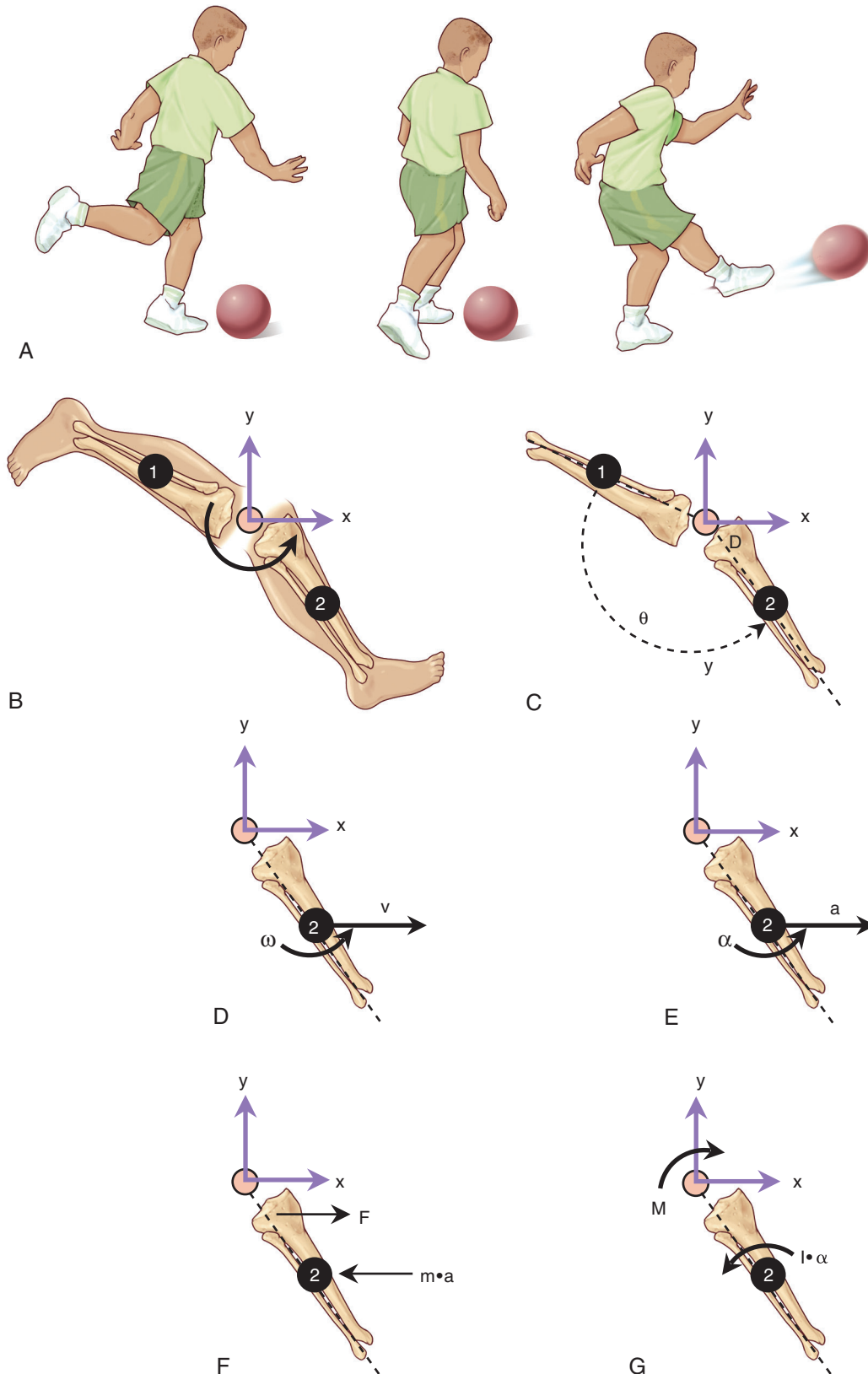


Fig. 2.1 (A) Sequence of leg positions while kicking a ball. (B) The motion of the leg in the acceleration phase of kicking in the sagittal plane at positions 1 and 2. (C) Displacement (D) and rotation (θ) of the tibia during the motion. (D) The linear (v) and angular (ω) velocity vectors. (E) The linear (a) and angular (α) acceleration vectors in the acceleration phase of kicking. (F) The linear and angular inertial forces, with the applied force (F) resisted by the inertial force ($m \cdot a$). (G) The applied moment (M) resisted by the angular inertia ($I \cdot \alpha$).

of the individual body segments to achieve stability or a particular motion. Gymnasts and divers use this concept to achieve multiple somersaults while in the air by tucking in the head and limbs closer to the center of the body, which is an example of angular acceleration due to changes in angular inertia (see Fig. 2.1G).

Newton's Third Law (Action-Reaction)

Newton's third law states that for every action there is an equal but opposite reaction. This law explains the idea that if a person pushes against a wall, it will in essence push back. Forces of action and reaction are equal in magnitude, but in the opposite direction. This concept is important in examining the principle of equilibrium and when using tools that assess forces being applied to the body of interest. Reaction forces act to constrain motions by reacting to an applied force. A daily application of this concept is simple walking and running. Every time a person places a foot on the ground, a force is exerted from the foot on the ground throughout the gait cycle. Simultaneously, however, a force of equal magnitude is exerted in the opposite direction from the ground up to the foot, which is termed a *ground reaction force*. Ground reaction forces are the forces applied by the ground because of the weight of the body. Similarly, within joints, muscles and connective tissues are found that create a *joint reaction force* (Fig. 2.2). Both passive and active stabilizers maintain the humeral head within the glenoid fossa of the glenohumeral joint,¹ which has implications for surgical repair of Bankart lesions by changing the arc of the glenoid and tendon transfers in changing the line of muscle action. If these joint structure changes alter the natural forces and/or the direction of these forces, joint instability would result, potentially leading to long-term degenerative changes.

When a body is balanced by equal but opposite reactions for every action, the system is considered to be in a condition of equilibrium, which means that the net effect of the applied forces is zero as well as the net moment about any point. Although the sum of the force and moment vectors is zero, the body may still be moving at a constant velocity. Assuming a state of static

equilibrium with a set of known force vectors, it is possible to determine unknown forces within a system. To do so, each force vector can be resolved into its individual components, such as the vertical (F_y) and horizontal (F_x) components for a two-dimensional coordinate system (Figs. 2.3 and 2.4D). Therefore the resulting equations for force and moment equilibrium in three dimensions are:

$$\Sigma F_x = 0; \Sigma M_x = 0$$

$$\Sigma F_y = 0; \Sigma M_y = 0$$

$$\Sigma F_z = 0; \Sigma M_z = 0$$

For static equilibrium, no linear or rotational motion occurs.

STATICS

Using the aforementioned equations of static equilibrium, static analyses evaluate the external effects of forces on a rigid body at rest or during motion with a constant velocity. Applied to the body, static analyses are used to further determine the magnitude and direction of forces at joints and in the muscles. Forces provide both mobility and stability to the body but also introduce the potential to deform and injure the body. Typically, healthy tissues are able to withstand changes in their shape, but a tissue structure that has been injured by disease or trauma may not be able to adequately sustain the same loads required to perform activities of daily living. To perform static analyses, we must be able to

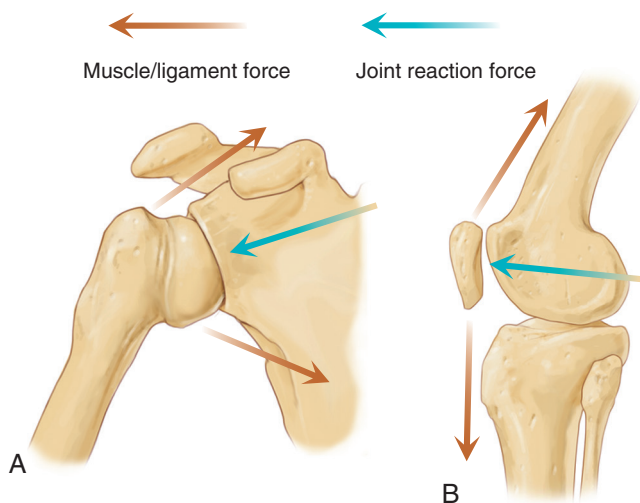


Fig. 2.2 Resulting joint reaction force at the glenohumeral (A) and patellofemoral (B) joints.

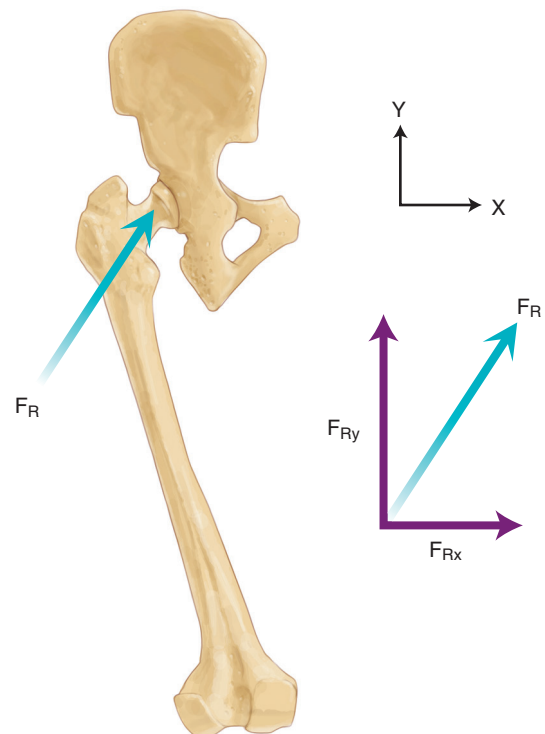


Fig. 2.3 Force vector (F_R) representing the joint reaction force between the femoral head and acetabulum, and F_R resolved into its individual component forces (F_{Rx} and F_{Ry}) in the x and y directions.

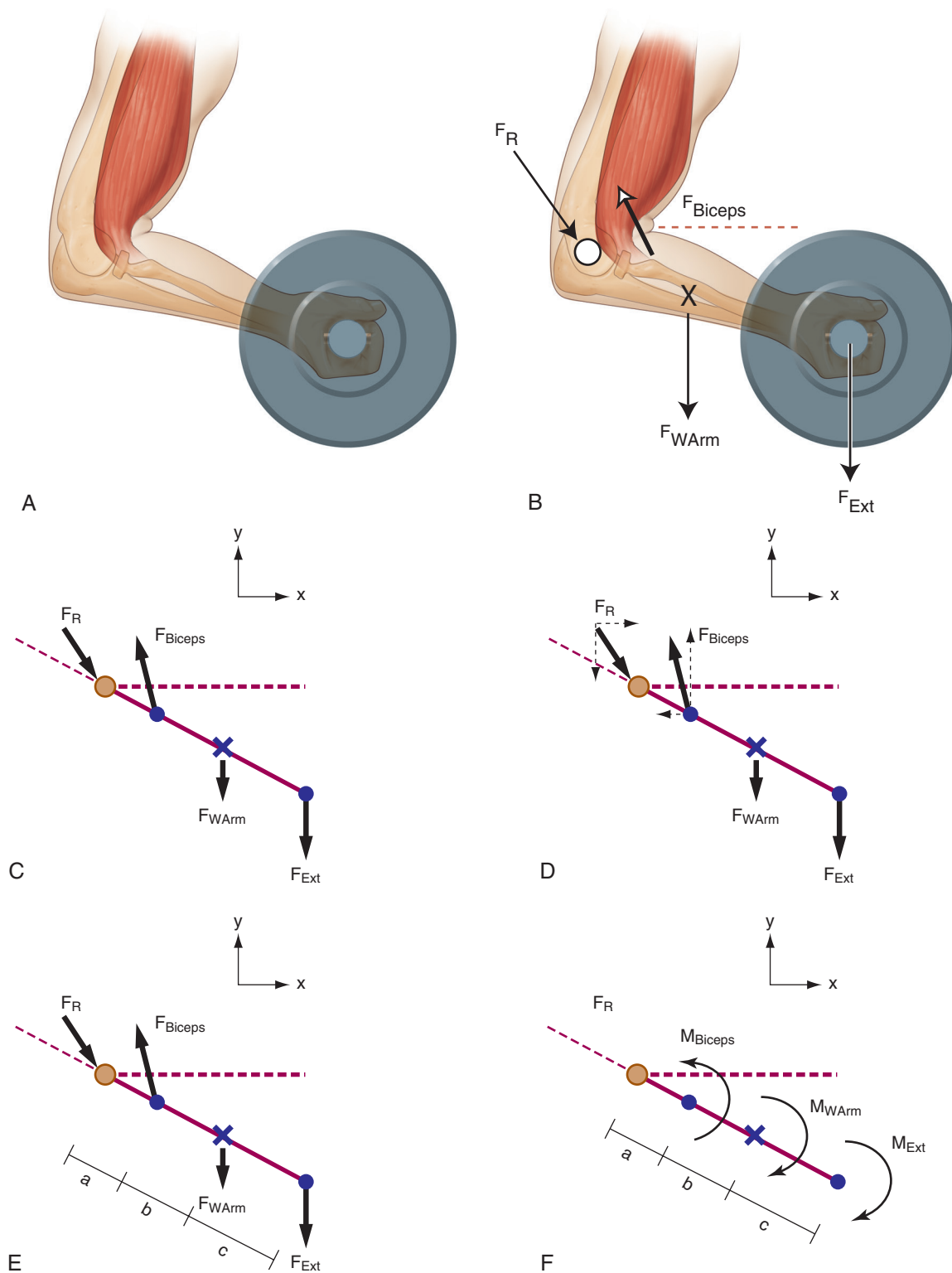


Fig. 2.4 (A) Simulation of a person performing a biceps curl. (B) The dumbbell applies an external force (F_{Ext}) downward in addition to the downward force due to the weight of the arm (F_{WArm}). The biceps muscle generates a force (F_{Biceps}) to the forearm and causes a joint reaction force (F_R) at the elbow to keep the joint stabilized. (C) A free-body diagram of the forearm representing each force as an arrow, with the head of the arrow pointing in the direction of the applied force. (D) These forces are then decomposed into component vectors. (E) The varying moment arms for each force vector are identified with respect to the axis of rotation at the elbow. (F) The biceps muscle creates a counterclockwise moment (M_{Biceps}) to resist the clockwise moments due to the weight of the arm (M_{WArm}) and dumbbell (M_{Ext}).

represent the complex interactions of forces and moments acting on a body through the use of vectors and free-body diagrams.

Force Vectors

The most common vector quantity in mechanical systems is a force. A force is applied on an object to create either a pushing or pulling response. Depending on the original state of the object, a force can cause a stationary object to move or alter the state of an object already in motion. Such forces can be either internal or external. Internal forces include those that hold a rigid body together, such as muscle tension generated within an extremity, whereas external forces are those applied to a rigid body. An example of an external force is the weight of an object being held in a person's hand. Forces can be further categorized into two subgroups: contact (tension, friction, external, and internal) and distance (gravitational and magnetic) forces. Distance forces act at a distance from the object or body with no direct physical contact. Additionally, forces can contain components acting in both the normal (perpendicular) and tangential (parallel) directions of the surface to which they are being applied.

Moment/Torque Vectors

Vector quantities can represent not only translational motions in response to an applied force but also the rotation, twisting, and bending of an object. A moment (or torque) is determined by the magnitude of the force acting about a point and the length of the shortest distance between the point and line of action of the force. This distance is known as the *moment* or *lever arm*. A larger moment arm requires less force to achieve equivalent angular motion about the axis of rotation, which means that if the force remains the same but the lever arm increases, the moment will be of greater magnitude. Although a moment can realistically be calculated about any point, typically it is calculated about a joint axis of rotation during biomechanical analyses. For example, when tension is generated in the biceps muscle during a biceps curl exercise, the tendon pulls on the forearm at a distance from the elbow axis of rotation (see Fig. 2.4). This tension from the biceps muscle generates a positive moment about the elbow that can either help maintain the posture of the elbow or increase the amount of flexion, depending on whether the biceps muscle force is equal to or greater than the applied external forces of the weight.

Free-Body Diagrams

To better evaluate a biomechanical system, such as forces being applied to a specified part of the body, free-body diagrams are an effective tool to simplify a complex analysis. Free-body diagrams allow for visualization and ease of calculation by properly identifying all the forces and moments acting on the body of interest in order to successfully achieve equilibrium. These diagrams are used by first drawing the body of interest and then isolating the body from its environment, only including the forces acting on the body. Again using the example of a biceps curl exercise to evaluate the elbow joint, the system should be drawn as only the forearm (with the radius and ulna combined as one), because this segment is the body segment of interest, with the elbow as the axis of rotation (see Fig. 2.4B). It is important to

note the position and orientation of the object by defining a coordinate system.

Applied forces are then identified and considered. Arrows representing the force vectors are drawn at every point where two (or more) bodies interact or join (see Fig. 2.4C). Recalling that internal forces are those that hold a rigid body together, two examples within the human body are muscle contractions and bony contact at a joint line. Internal forces are further examined with the concept of stress, which is introduced in a later section of the chapter. External forces can be the weight of an object, friction, or gravity applied at the center of mass of a body segment. In a simplified situation, flexion of the elbow is counteracted by the weight of the forearm itself and the weight of the dumbbell. The point of application of the forearm weight is at the segment's center of gravity. Commonly this point is determined from anthropometric data. The point of application of the biceps force on the forearm is its tendon insertion. The remaining force is the result of contact with the distal end of the humerus and the proximal end of the forearm, whereby the point of contact is approximately at the joint surface of the bones. These forces can then be further resolved into their respective components along the previously identified x- and y-axes (see Fig. 2.4D).

In addition to forces, the forearm experiences various applied moments. Because the points of application for the forces are known, the distances of each force along the forearm from the axis of rotation—the elbow—is also known (see Fig. 2.4E). Using these distances as moment arm values, the next step is to identify moments created by each force. The direction of the moment is relative to the applied force and its relationship to the axis of rotation. No moments are generated by the forces at the elbow because their moment arms are zero (see Fig. 2.4F). Once all the forces and moments applied to the body or body segment of interest are identified, they can be summed to determine the resultant force and moment. When the body is in a state of static equilibrium, the resultant forces and moments sum to zero, according to Newton's third law of motion.

Ligament and Joint Contact Forces

These concepts of static analyses can be applied not only to whole-body analyses but also at the joint and tissue level. For the typical joint, forces can be related to compression and shear. For example, when a person is standing with the knee in full extension, the tibial plateau and femoral condyles experience compressive forces in the normal direction to each articular surface. However, an increased shear force is experienced during an anterior drawer test or when stopping quickly in a sporting event. Shear forces are experienced in the tangential direction along the tibial plateau. These forces are not only experienced between the bony structures but also transmitted through the soft tissue structures. During an anterior drawer test or quick stop, the ACL can become significantly loaded to resist anterior tibial translation and provide stability at the joint. The ACL becomes loaded as a result of a generated tensile force within the ligament to maintain equilibrium. However, after an injury such as an ACL rupture, opposing shear forces cannot be transferred through the ligament, and thus equilibrium cannot be

maintained at the joint without excessive translation, resulting in anterior instability.

DYNAMICS

Based on the equations from Newton's second law of motion, dynamic analyses evaluate bodies in motion and can be divided into two subgroups: kinematics and kinetics. For dynamic systems, the forces and moments do not have a net value of zero, which violates the principle of equilibrium, and thus a different approach must be taken. Kinematics simply describes the motion of bodies, without regard to the factors that cause or affect the motion, by characterizing the geometric and time-dependent aspects of the motion. Conversely, kinetics is based on kinematics but includes the effects of forces and moments. Motion analyses and sports mechanics typically involve dynamic systems.

Kinematics

Simply stated, kinematics deals with motions without regard to forces and moments. These motions include translations and rotations. Translations are simply the linear motions in which all the parts of a rigid body move simultaneously in the same direction as every other point in that body and at the same velocity. Rotations are the angular motions of a rigid body along a circular path and about an axis of rotation. During passive knee flexion, the tibiofemoral joint undergoes both linear and angular motions. As the knee progresses through the ROM, the tibia experiences rotation in the sagittal plane, and simultaneously, the point of contact between the tibial and femoral articular surfaces translates in the posterior direction. Although flexion is the primary angular motion, internal tibial rotation also occurs, which exemplifies the complex motion that can occur at a single joint.

Linear and Angular Kinematics

The spatial components used to describe linear kinematics are position, distance, and displacement. Position is a vector that simply defines the location of the object in space relative to a reference frame, commonly the center of a joint or a point of contact. Once the initial position changes, the object has moved a specified distance and displacement. Distance is a scalar quantity that describes the length of the entire path traversed. Displacement is a vector quantity (d) defining the shortest distance (straight line) between the starting and ending positions of the object independent of the path taken (see Fig. 2.1C). Additionally, temporal descriptions of linear kinematics are speed, velocity, and acceleration. Velocity is a vector quantity (v) describing the rate of change of the position with respect to time, whereas speed is a scalar quantity equal to the magnitude of the velocity vector (see Fig. 2.1D). Furthermore, acceleration is also a vector quantity (a) that describes the rate of change of velocity with respect to time, but it is commonly used to express an increase or decrease in speed (see Fig. 2.1E).

Angular kinematics is experienced when a change in angular position occurs such that the body undergoes rotational motion about an axis of rotation. The angular distance through which the body moves is equal to the length of the angular path. Angular

kinematics can describe rotatory motion for a body segment, such as the lower leg during a kick (see Fig. 2.1A), in addition to the body as a whole, such as for a gymnast swinging on the bar to perform a giant circle. Descriptors used for these motions with respect to time are *angular position*, *displacement*, *distance*, *velocity*, and *acceleration*. These terms are similar to those used for linear kinematics, but they apply to a rotatory motion (see Fig. 2.1C–E). The lower leg undergoes large changes in angular velocity (ω) and acceleration (α) throughout the ROM for activities such as running, swimming, or kicking a soccer ball. Most general movements in the body are a combination of both linear and angular motion. For example, during a normal gait cycle, the lower extremities translate and rotate. Similarly, during athletics, combined motions are clearly illustrated during pitching, as the glenohumeral joint creates an instantaneous axis of rotation when the ball is swung along an imaginary arc about the shoulder in addition to being propelled forward.

Relative Motion

When considering forces and moments applied to a body, it is important to know the relative relationship in terms of both position and orientation for the individual bodies to each other in addition to the overall environment. An illustration of this concept is that a moving object may appear to have a different motion for two different observers depending on their location to the moving object. Knowing relative relations allows for an accurate description of the system when solving for unknown forces by establishing a frame of reference, or coordinate system. If a frame of reference is not identified, the measurements become irrelevant. A frame of reference describes the position of one body with respect to another whereby a relative measurement can be performed. This measurement is made by comparing the change in position and orientation of the object to the reference frame. For example, the position and orientation of the femur can be defined relative to the pelvis when describing motions of the hip or of the tibia with respect to the femur during kicking (see Fig. 2.1B).

Joint Motions

Motion at an articular surface can be described in terms of three motions that exist as a result of convex surfaces moving on a concave surface (Fig. 2.5). A *rolling motion* occurs when the bone with the convex surface rotates to cause a change in the point of contact for both articular surfaces in addition to a corresponding linear motion. A ball experiences this type of motion when it is rolled across a smooth surface. A *sliding motion* is experienced when one articular surface translates across the other with no rotation and progressively changes the point of contact. Sliding occurs when a box is pushed across a smooth surface. A *spinning motion* occurs when there is a single point of contact on the fixed surface and the point of contact changes on the rotating surface that does not undergo any linear motion. Spinning can be experienced on an automobile tire when the tire is rotating but the automobile is not moving forward or backward because of ice on the road. For certain joints, a combination of two or all three of these motions occurs. The motion achieved at a joint is based on the shape of the articular surface.

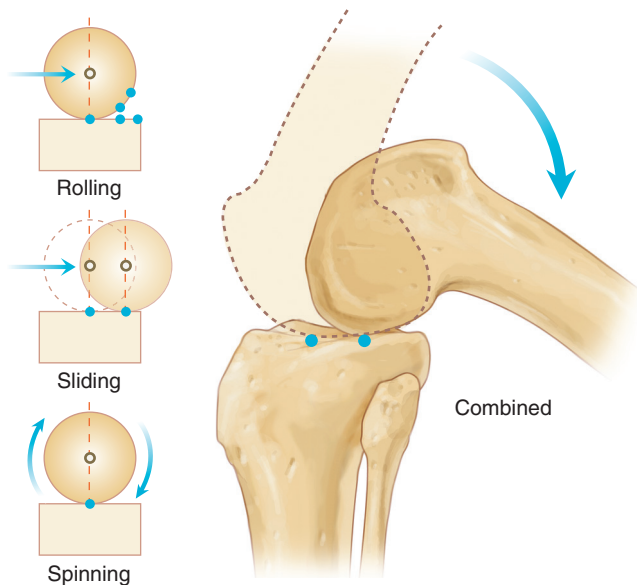


Fig. 2.5 Three fundamental motions that occur between articular surfaces. The point of contact changes on both articular surfaces during a rolling motion. The point of contact on the moving surface remains constant during a sliding motion. A single point of contact occurs on the fixed surface during a spinning motion. Some joints, such as the tibiofemoral joint, experience up to all three of these motions simultaneously.

For example, at the tibiofemoral joint during flexion, the knee experiences femoral condylar rollback on the tibial plateau. This phenomenon encompasses all three motions (rolling, sliding, and spinning) through flexion rotation, posterior translation, and external rotation, respectively, of the femur relative to the tibia. Although it is important and beneficial to understand the translations and rotations occurring at a joint, it can also be beneficial to understand the forces and moments associated with the observed translations and rotations, thus introducing the necessity of kinetic analyses.

Kinetics

Kinetics is the branch of mechanics that describes the effect of forces and moments on the body by utilizing Newton's second law of motion. More specifically, by using kinetics, it is possible to determine the forces and moments on a joint produced by mass, muscle tension, soft tissue loading, and externally applied loads. The appropriate range of joint loading that can be considered noninjurious can also be determined using kinetic analyses. Conversely, situations can be identified that produce excessively high moments and forces that exceed these limits, thus leading to musculoskeletal injuries. If an excessive valgus moment is experienced at the knee because of contact during a sport such as football, the medial collateral ligament may become overloaded and rupture in an attempt to maintain stability and resist the applied load. High forces can also be applied internally by overloading tendons as a result of extreme muscle tension.

CLASSES OF MECHANICS

Mechanics is the study of forces and their effects on motion. Depending on the system being studied, different branches of

mechanics may apply. When examining the behavior of solid bodies, such as the forces acting on human limbs and their motions, rigid body mechanics may apply. When examining the effects of a body moving through air or water, fluid mechanics are applicable.

Rigid Body Mechanics

In rigid body mechanics, it is assumed that any deformation caused by forces acting on a body is negligible. By assuming that a body is nondeformable—that the distance between any two points in the body will remain constant for any given translation or rotation—any changes in the shape of the body due to an applied force can be ignored, making kinematic analyses much simpler. No material in the human body can truly be considered a rigid body because all tissues undergo some degree of deformation, and thus it is important to note when the rigid body assumption is applicable. If the deformations experienced by a body are much smaller than the translations or rotations of that body, then the rigid-body principle can be applied. For example, when analyzing gait, the translations and rotations of the lower extremity will be much greater than any deformations experienced by the segments of the lower extremity, allowing each segment to be treated as a rigid body. Similarly, when using training equipment such as during weight lifting (see Fig. 2.4), the bar and limb segments of the upper extremity can be considered rigid bodies to allow for kinematic and kinetic analyses because any translations or rotations are much greater than any deformation in the equipment or limbs. Additionally, if one material is much stiffer than another, the stiffer material can be considered to be a rigid body that undergoes no deformation. For example, when performing mechanical testing of a joint complex, such as the femur–medial collateral ligament–tibia complex, the bones can be considered rigid bodies because they are much stiffer than the ligament tissue.

Fluid Mechanics

Fluid mechanics takes the concepts of statics and dynamics for solid bodies and applies them to fluids in order to study their behavior under applied forces and displacements. The term *fluid* refers to both gases and liquids and can be defined as a material that flows freely as the result of the application of a force and cannot support shear stresses. The most important parameter describing the behavior of a fluid is its viscosity (μ), a measure of how a fluid resists flow because of an applied force. Certain fluids, such as blood, have higher viscosities, whereas other fluids, such as air, have very low viscosities that can be considered to have negligible resistance. The viscosity of blood depends on its cellular content; when hematocrit levels rise above normal, blood viscosity rises and flow becomes more resistant to the forces pumping it through the body, leading to higher blood pressure as the heart works harder to pump blood through the body. During respiration, viscosity contributes to airflow patterns through the trachea and bronchi, relating to resistance of airflow. Although the viscosity of the air does not change, changes in the airway diameters due to disease can increase resistance to airflow, leading to respiratory pathology.

Additionally, movement of a solid object through fluid can result in the phenomenon of “drag,” or fluid resistance, which reduces the velocity of the body. Drag is dependent on both the surface area exposed to the fluid in the direction of movement and on the velocity of the object: the greater the velocity and surface area, the greater the fluid resistance. At the whole-body level, water resists the movement of the body while swimming, and air resists the movement of sprinters traveling at high speeds. To reduce the effects of drag, athletes position their bodies during running or sprinting to reduce the surface area of their bodies perpendicular to the direction of motion.

Fluids can be considered to be either Newtonian or non-Newtonian. Newtonian fluids, such as water or air, exhibit a linear relationship between force applied to the fluid and the fluid deformation rate. Non-Newtonian fluids do not exhibit this relationship, an example being blood, as a result of its composition of particles (cells) suspended in aqueous plasma. The importance of Newtonian versus non-Newtonian flow relates to being able to accurately predict the behavior of fluids in the body, particularly blood. Depending on the size of the vessel, Newtonian blood flow may be assumed in some cases even if the fluid is non-Newtonian. The accurate prediction of blood flow patterns is important for understanding how abnormal blood flow can lead to atherosclerosis or low blood perfusion in tissues.

Viscoelasticity

A tissue is viscoelastic when it possesses to some degree both solid-like characteristics, such as elasticity and strength, and liquid-like characteristics, such as flow depending on temperature, time, rate, and amount of loading. Most tissues within the musculoskeletal system demonstrate at least some degree of viscoelasticity. After plotting the load-elongation results of a nondestructive tensile load, there remains a region between the loading and unloading curves that is known as hysteresis and clearly shows the time-dependent effects that viscoelasticity introduces (Fig. 2.6). Preconditioning with repeated loading and unloading of the tissue decreases this area of hysteresis and

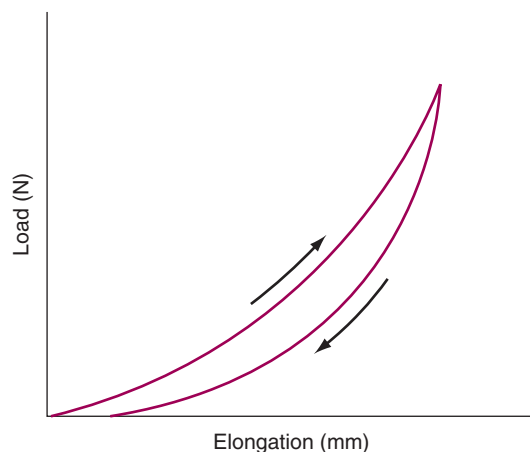


Fig. 2.6 Load-elongation curve of a biologic soft tissue in response to the application of a tensile load. The area between the loading and unloading curves represents the energy absorbed by the tissue during this loading regimen, or hysteresis.

maximizes elongation of the tissue, which is why athletes precondition the tissues in their bodies by completing repetitive stretching activities.

A viscoelastic material experiences the phenomena of creep and stress-relaxation. *Creep* describes a progressive increase in elongation of a material when exposed to a constant load over time (Fig. 2.7A). In simple terms, creep can be considered the tendency of a material to move or deform in response to a constant stress. Conversely, with stress-relaxation, a decrease in load occurs over time upon application of a constant elongation (see Fig. 2.7B). Stress-relaxation is the phenomenon that occurs in a material to relieve stress under a constant strain due to the liquid-like characteristic of viscoelasticity. After a period of time, for both creep and stress-relaxation, the tissue will reach an equilibrium state of elongation and load, respectively.

A practical use of both stress-relaxation and creep in the clinic is for initial graft tensioning of ligament or tendon reconstructions. Over time it is impractical to expect that the initial graft tension will be maintained after being fixed. The length of the graft will inevitably increase from its original length because of a creep response. Moreover, the rate of loading is important because a faster loading rate will result in greater stiffness. Viscoelasticity is also experienced in the articular cartilage of the knee. For example, as the rate of compressive loading increases, such as during running activities, the tissue becomes much stiffer.² This increase in stiffness allows for improved protection to the underlying bone while forces at the joint are high. Overall there is not a substantial strain rate dependency on tendons and ligaments; however, bones exhibit large changes in stiffness with increases in the rate of loading. In addition, bone has a greater compressive strength than tensile strength, which is advantageous during sporting activities when a sudden powerful blow may occur to a portion of the body.

BIOMECHANICAL METHODS

In traditional approaches to evaluating both static and dynamic systems, it is assumed that each object in the system is a rigid body. Biomechanics accounts for the biologic materials of the musculoskeletal system in a more realistic manner by considering that load is experienced at the tissue level and that these tissues are deformable. These biologic materials include both soft (articular cartilage, tendons, ligaments, capsular tissues) and hard (bone) tissues. Tissue structures that have been weakened by disease or trauma may not be able to adequately resist the loads that are applied.

Structural Properties

Structural properties are represented by the mechanics of a structure that may incorporate multiple materials or tissue types, including the response of this complex to tensile, shear, and compressive loading. For example, the structural properties of the femur-ACL-tibia complex can be determined in response to a tensile load to assess its load-elongation behavior. To do this, a tensile force (F) is applied to the bone-ligament-bone complex, causing the tissue to become stretched until the complex ruptures. While loading is applied, the corresponding increase of length

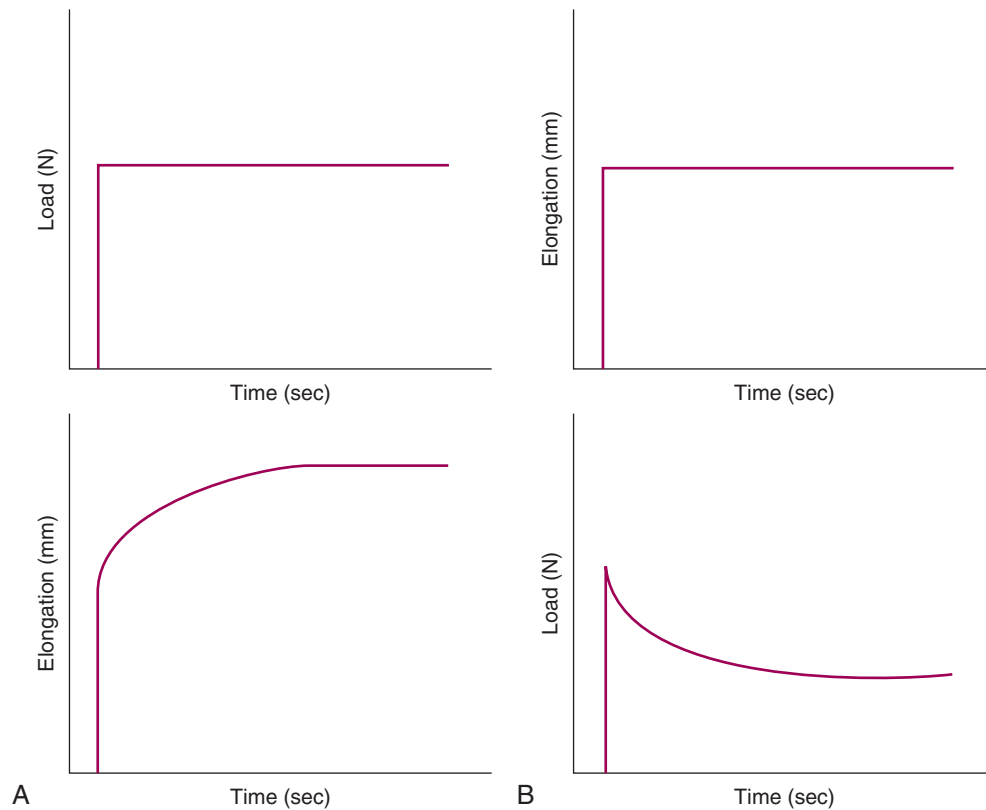


Fig. 2.7 Viscoelastic phenomena exhibited by biologic tissues include creep (A), that is, response due to a constant applied load, and stress-relaxation (B), that is, response due to a constant applied elongation over some time.

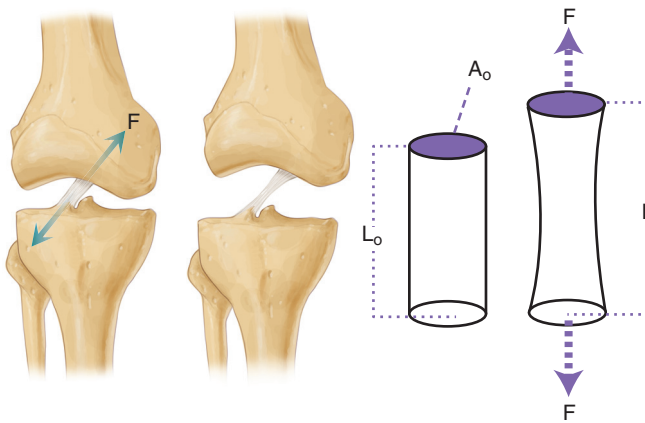


Fig. 2.8 Tensile loading along the longitudinal axis of the anterior cruciate ligament causes a change in length from its initial length (L_0) to an elongated state (L). A_0 , Cross-sectional area; F , applied force.

is measured (Fig. 2.8). The resulting nonlinear load-elongation curve that is typical of biologic soft tissues consists of four primary regions (Fig. 2.9A). As the tensile load is first applied, the relationship between load and elongation is nonlinear and is referred to as the *toe* region. This level of loading is typically experienced during a clinical examination as normal fiber recruitment occurs. However, with increasing loads, the relationship becomes more linear and represents loading experienced during daily and

sporting activities. The slope of this linear region of the curve is known as the *stiffness*. When the soft tissue begins to sustain more load than the structure can support, plastic deformation begins to occur. Finally, the entire structure approaches the failure region and completely ruptures. The point at which failure occurs is known as the *ultimate load*. Another structural property is the energy absorbed as a result of loading of the tissue, which is determined by the area under the load-elongation curve.

Mechanical Properties

It is also important to understand the mechanical response of an individual tissue or material, which is independent of specimen geometry, specifically of the cross-sectional area and initial length, by using normalized load and deformation parameters. Mechanical properties can be used to evaluate the quality of the tissue when making comparisons between normal, injured, and healing states and are represented by the stress-strain relationship. *Stress* is defined as the amount of force applied per unit area and is one of the most basic engineering principles. *Strain* can be considered a dimensionless measure of the degree of deformation and is defined as the change in length per unit length. Specimens with a greater length can withstand more total deformation, whereas specimens with a larger cross-sectional area can carry loads of greater magnitude. The mechanical properties can be derived from a plot of the stress and strain data and may include modulus, ultimate strength, ultimate strain, and strain energy density. Stress-strain relationships are obtained

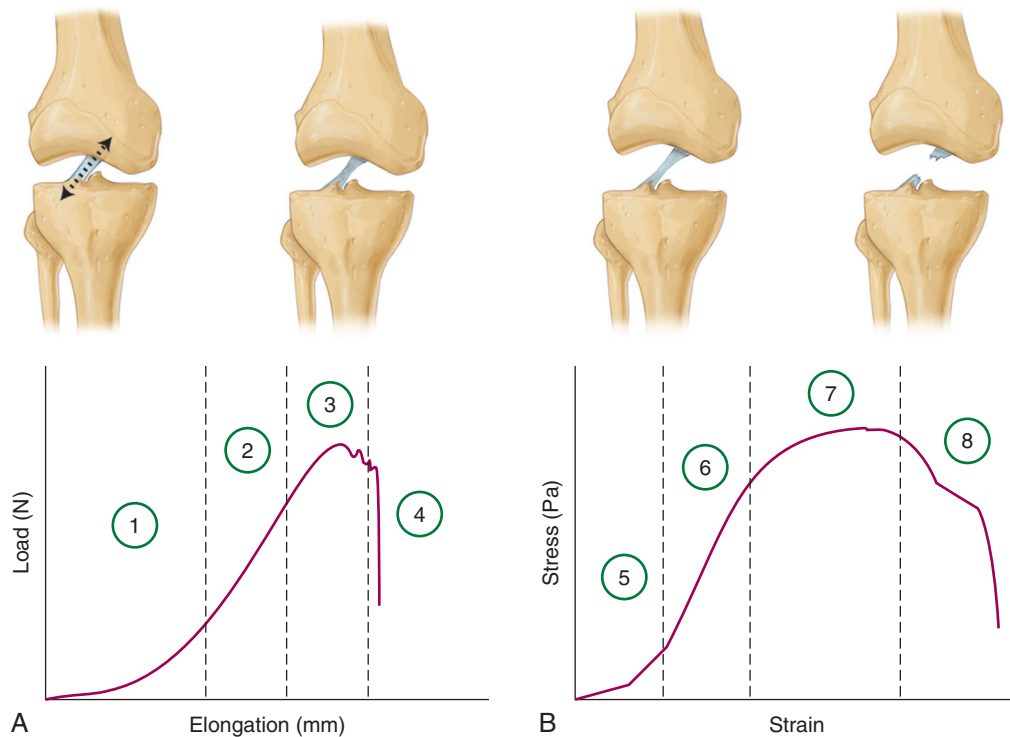


Fig. 2.9 (A) Load-elongation curve of a bone-ligament-bone complex in response to tensile loading characterizing its structural properties. Regions 1, 2, 3, and 4 correspond to the toe region, linear region, partial failure of the complex, and complete rupture of the complex, respectively. The loading of the tensile-bearing structure during activities of daily living will remain in regions 1 and 2. (B) Stress-strain curve of the ligament substance that characterizes its mechanical properties. Regions 5, 6, 7, and 8 correspond to the toe region, linear region, partial failure of the material, and complete rupture of the material, respectively. Pa, Pascal.

experimentally during tensile, compressive, or shear loading of the excised tissue.

There are four distinct regions along a stress-strain curve for biologic tissues (see Fig. 2.9B). A typical stress-strain curve begins with a nonlinear toe region (region 5). Stretching of the crimped collagen fibrils occurs within this region as the fibers are being drawn taut before significant tension can be measured. Strain becomes linearly proportional to stress in region 6, and the slope of the curve in this region can be calculated to determine the tangent modulus of the tissue. The area under the curve within this region can be referred to as the *strain energy density*. The tissue returns to its original length or shape once the stress is removed within this zone, which is usually reached during most daily activities. Furthermore, the energy used to deform the tissue is released when the applied stress is removed. When the tissue experiences extreme and abnormally large strain, the tissue undergoes only a marginal increase in corresponding stress (region 7). It is at this point that the tissue begins experiencing microscopic failures. The area under this region of the curve represents plastic deformation energy. Once the tissue undergoes this amount of deformation, the tissue does not recover and return to its original state in its entirety upon release of the deforming stress. If the tissue continues to deform, it will eventually experience complete failure (region 8). The mechanical properties of a tissue can be used to evaluate injured and healing states. When ligament repair is being evaluated, the medial collateral

ligament has much lower modulus and maximum stress mechanical properties during the healing process after rupture, although it shows some improvement over time without full recovery.³

To determine the mechanical properties of a knee cruciate ligament, a tensile load is applied along the long axis of the excised tissue sample after measuring the original cross-sectional area of the tissue (see Fig. 2.8). When a load is applied in the direction perpendicular to the tissue structure such that there is only a linear change in deformation, the load and deformation are considered to be axial stress and strain, respectively. As the load is applied along the long axis of the ACL in Fig. 2.8, a decrease in the cross-sectional area at the midsubstance occurs in response to an increase in the overall length (L) of the ligament from the initial length (L₀). This response is important to consider in converting load data, a structural property, to stress data, a mechanical property, because the structure is changing dimensions throughout the testing. In calculating the mechanical properties, these changes are often ignored and the original, initial conditions are used. Compressive loading is commonly used to test bone or articular cartilage, whereas tensile and shear testing may be used for connective structures (e.g., tendons, ligaments, and capsular tissues).

Shear stress and strain act parallel to the surface of the tissue (Fig. 2.10), and the body is observed to deform into a rhomboid with sides equal to 1 if the body is originally a unit cube. Shear strain can be measured as the angular change (γ) at any point

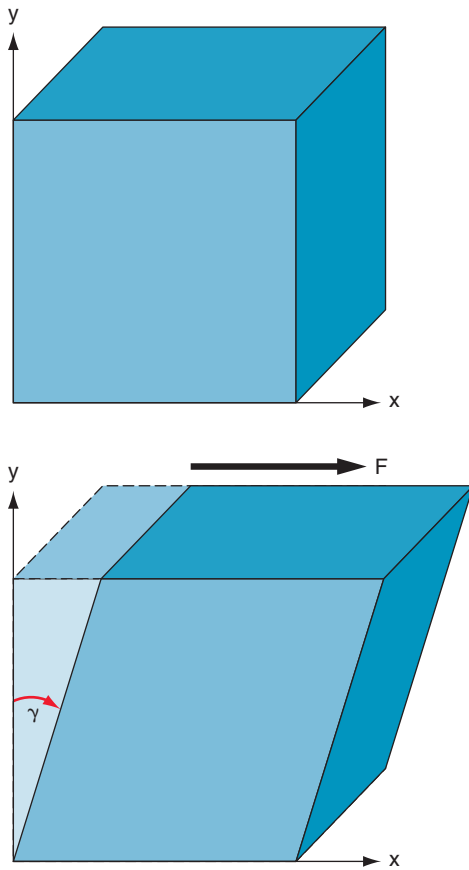


Fig. 2.10 Applying a shear force (F) to the top of a cube results in deformation from the y -axis through an angle, γ , in the direction of the load.

in the body owing to the applied shear force (F). Experimental shear loading of knee articular cartilage has shown that loading parallel to the articular cartilage surface may lead to the degeneration of cartilage and osteoarthritis.⁴

Factors Affecting Biomechanical Properties

Structural and mechanical properties can be significantly affected by both biologic and testing-related factors. The age and activity level of the person from whom the specimen was acquired are examples of biologic factors, and tissue storage and gripping are examples of testing-related factors. As a child or adolescent matures toward skeletal maturity, tendon and ligament tissue modulus increases slowly, but the maximum stiffness rapidly increases because of increasing tissue size.⁵ In addition, the failure mode is predominantly through the bone attachment rather than in the midsubstance in growth and development.⁶ However, after skeletal maturity, both modulus and maximum strain decrease two- and threefold, respectively, up to the age of 50 years, and the primary failure mode changes to midsubstance rupture. Beyond age 50 years, the primary failure mode is bony avulsion.⁷ Donor activity level can have a significant biologic effect on the structural and mechanical properties. Tissue samples obtained from donors who have been immobilized or inactive for an extended period possess tissue stiffness and strength that is significantly less than that for control samples.^{8,9} These factors

can influence the structural and mechanical properties and must be taken into account when performing an experiment.

Although testing-related factors can also influence the final structural and mechanical properties, these factors are controllable.

Tissue Storage

Improper tissue storage can significantly degrade the tissue's properties. To ensure that the tested tissue better represents the *in vivo* tissue, the tissue should be stored below -15°C for a minimal duration, and the number of freeze-thaw cycles should be kept at a minimum. These conditions help slow the release of proteinases from the cells, thus maintaining the extracellular matrix. The temperature at which the test is performed will also influence the structural and mechanical properties. Ligaments are stiffer at room temperature compared with body temperature.

Testing Rate

The rate at which the tissue is strained will alter the tissue's properties. Ligaments and tendons are mildly rate-sensitive, producing only a 10% to 15% increase in stiffness for a $\times 100$ increase in strain rate.¹⁰ By contrast, bone is more rate-sensitive, showing greater than 25% increase in stiffness for a $\times 100$ increase in strain rate.¹¹

Stiffness Versus Flexibility Testing

To examine the functional role of individual joint structures, researchers have used stiffness- and flexibility-based testing methods. For stiffness-based testing, a displacement (either translations, rotations, or a combination) is applied to the joint and the resulting forces and moments are recorded. The structure of interest is removed, and the same displacement is applied to the joint. The change in forces and moments is attributed to the structure that was removed. In flexibility-based testing, however, known forces, moments, or a combination are applied to the joint, and the resulting displacements are recorded. Once again, the structure of interest is removed, and the same force-moment profile is applied. The displacement change is attributed to the structure of interest. These two tests provide valuable information about the role of individual structures such as the forces and moments developed during activities of daily living (stiffness testing) or the structure responsible for restraining excessive motions (flexibility testing). A disadvantage of the flexibility testing method is that it is sequence-dependent, meaning that the role of each structure is influenced by the structures that have previously been cut. For the stiffness-based testing, the contribution of an individual structure can be overestimated if this structure physically interacts with adjacent tissues. An example is the wrapping of the ACL and posterior cruciate ligament as the knee extends. If the ACL is removed, the forces and moments provided by the posterior cruciate ligament also change because the constraint provided by the ACL is now gone. The change in forces and moments is greater than what is truly generated by the ACL.

Practical Application

The concepts presented throughout this chapter provide a way of understanding and quantifying how the human body operates

to perform activities of daily living. This information can be used in orthopedic sports medicine to improve athletic performance, prevent injury, and develop surgical reconstruction techniques.

Improving Performance

Endurance is an important part of performance for any athletic activity. As activity is sustained over time, energy stores within the body are depleted, leading to fatigue. To postpone fatigue, energy expended in the activity should be minimized. For example, proper posture in cycling decreases the air resistance that must be overcome by the cyclist. Another example involves energy recuperation through the use of muscle and tendon elastic energy during running. As the foot of a runner strikes the ground, the force of the impact elastically elongates the muscles and tendons of the leg, allowing them to store energy to be released at toe-off. By properly preconditioning these tissues prior to activity, a greater proportion of impact energy can be stored, lessening the amount of active energy that must be expended by the runner's legs to achieve the same performance. Biomechanical analysis is also useful for maximizing force output, as in striking a baseball with a bat or throwing a football. As coaches and trainers are well aware, the relative positioning of each body segment is extremely important in generating the maximum moments between segments that will translate to physical power, or the rate of doing work.

Injury Prevention

Understanding the normal forces and moments or displacements applied to different structures surrounding joints—as well as their structural and mechanical properties—can be powerful in preventing injuries. In addition, analyses of tissue failure can allow clinicians and researchers to develop devices and training protocols to minimize injury risk. A high rate of collateral knee ligament injury among football linemen was a driver for the development of protective knee braces. Female athletes' greater risk of ACL rupture has led to the development of muscle training programs designed to provide better dynamic stabilization and to reduce the number of ACL ruptures. Additionally, normal biomechanical data are also useful during the creation of rehabilitation protocols. A rehabilitation protocol that maintains ROM and muscle strength is advantageous to the long-term healing of the patient. However, overaggressive rehabilitation can lead to failure of the repair. Knowledge of normal biomechanical properties can be used to design a regimen that maintains motion and muscle tone while also protecting the susceptible repair.

Surgical Reconstruction

The purpose of surgical reconstruction is to restore function by replacing nonfunctional tissue with a graft or other implant. However, to adequately restore function, it is important to understand the mechanics and biomechanical properties of the original tissues and whatever material is being used to replace them. Any materials being used to replace natural tissues, such as tendons or ligaments, must be able to function under the same ROM experienced by those connective tissues while also

maintaining adequate strength and stiffness. For example, the ACL is part of a complex loading environment experiencing both shear and longitudinal stresses during normal gait; it must resist anterior translation of the tibia with respect to the femur. The graft choice for ACL reconstruction must match the stiffness of the original tissue to sufficiently resist anterior translation of the tibia for the forces and moments experienced during gait. However, it must also be compliant enough to allow for the normal range of knee motion. Time-dependent viscoelastic properties of the graft must also be taken into account because any amount of creep occurring after the surgical reconstruction can result in increased translations at the joint. Similarly, for acromioclavicular joint reconstruction after shoulder separation, the structural properties of the graft construct must match the stiffness of the acromioclavicular and coracoclavicular ligaments to provide stability for the scapula without restricting normal ROM or failing under normal loading conditions for the shoulder.

Functional Tissue Engineering

Although researchers have developed novel tissue engineering therapies since the late 1980s, surgeons remain concerned about the fragility of tissue-engineered constructs (TECs) and their efficacy in patients after surgery.¹² In 2000, “functional tissue engineering” (FTE) was advanced, so that engineers, biologists, and surgeons might develop design criteria to judge potential TECs. Central to FTE was a set of core principles,^{13,14} including the need to “establish in vivo load and deformation histories for native tissues under a wide range of activities of daily living (ADLs),” “establish safety factors and relevant biomechanical properties for native tissues,” and compare the properties of repairs with similar histories for native tissues. FTE has been applied to the repair of numerous load-bearing tissues, including articular cartilage¹⁵ and tendon.^{16,17} Researchers have determined in vivo forces in the goat ACL and patellar tendon for numerous activities of daily living,¹⁸ as well as in the rabbit flexor profundus tendon, patellar tendon, and Achilles tendon.^{13,14,19–21} These studies revealed that although tendons can experience a range of in vivo forces up to 40% of failure force,²² ligaments such as the ACL routinely never exceed 7% to 10% of failure force.¹⁸ These studies established “functional” loading and deformation regions in which tissues normally reside for various ADLs.²³ Tissue engineers could thus provide a minimum set of design criteria for specific tissue applications by (1) knowing the peak in vivo forces for a tissue and adding a safety factor to account for still higher unexpected forces not normally experienced by the tissue and (2) knowing the tangent stiffness or slope of the force-deformation curve for the tissue during failure testing. Applying these two design criteria (exceeding peak in vivo forces with safety factor and matching tangent stiffness) has led to improved cell-collagen TECs through changes in cell and collagen density and through mechanical preconditioning of the constructs before surgery.^{23–26} This strategy allows for rapid determination of whether a repair is acceptable by one or both criteria or requires redesign. Parallel in vitro studies using TECs that contain cells from the same animals as those for repair assessment have also permitted the establishment of positive correlations between in vitro stiffness

and in vivo repair stiffness.²³ These correlations, in turn, have permitted optimization of the conditions that most enhance in vitro stiffness.²⁷ These new conditions can then be periodically evaluated to determine changes in repair outcome.

Additional studies in FTE are now under way to meet new challenges. Investigators will need to broaden their design criteria to include biologic and mechanical measures. This approach requires paradigms that focus on the factors (such as gene expression, protein synthesis, and collagen alignment) that greatly affect repair results. Furthermore, additional research is needed to more faithfully simulate ADLs for tissues that typically experience complex loading patterns (e.g., the cruciate ligaments).²⁸ Finally, the process of developing these cell-based therapies must be accelerated while controlling costs so that FTE can be used to treat patients in the near future.

CONCLUSIONS

This chapter introduced the basic terminology and concepts of statics, dynamics, and mechanics of materials using examples relevant to sports medicine. With these foundations, the reader should be able to communicate using common units of measure to describe various physical quantities. The chapter also presents concepts that illustrate how to approach both a static and dynamic system to calculate unknown forces and moments experienced on the body through use of a free-body diagram. A description of biologic materials within the musculoskeletal system is also presented. Although biomechanics has great depth and breadth, these concepts should allow the reader to begin making the link between sports medicine and biomechanics.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Abramowitch SD, Yagi M, Tsuda E, et al. The healing medial collateral ligament following a combined anterior cruciate and medial collateral ligament injury—a biomechanical study in a goat model. *J Orthop Res*. 2003;21:1124–1130.

Level of Evidence:

I

Summary:

Combined ACL reconstruction and MCL repair resulted in an ACL graft that lost force over time, overloading the MCL and producing an MCL with inferior biomechanical quality compared with an MCL after an isolated injury.

Citation:

Butler DL, Juncosa-Melvin N, Boivin GP, et al. Functional tissue engineering for tendon repair: a multidisciplinary strategy using mesenchymal stem cells, bioscaffolds, and mechanical stimulation. *J Orthop Res*. 2008;26:1–9.

Level of Evidence:

Review

Summary:

Through the principles of functional tissue engineering, tissue-engineered repairs have been created that produce tendon tissue that matched the normal tendon failure curve to 150% of the normal in vivo forces. During these studies, in vitro construct stiffness predicted final repair stiffness; however, additional biologic in vitro predictors of in vivo outcome are needed to continue advancing the field.

Citation:

Debski RE, Yamakawa S, Musahl V, et al. Use of robotic manipulators to study diarthrodial joint function. *J Biomech Eng*. 2017;139(2).

Level of Evidence:

Review

Summary:

An improved understanding of the structure and function of diarthrodial joints needs to be obtained to gain a better understanding of injury mechanisms and improve surgical procedures. Robotic testing systems have been developed to measure the resulting kinematics of diarthrodial joints as well as the in situ forces in ligaments and their replacement grafts in response to external loading conditions to achieve this purpose.

Citation:

Guilak F, Butler DL, Mooney D, et al. *Functional Tissue Engineering*. New York: Springer-Verlag; 2003:480.

Level of Evidence:

Book

Summary:

This book discusses the principles of functional tissue engineering and their application across different tissue systems.

Citation:

Nordin M, Frankel V, eds. *Basic Biomechanics of the Musculoskeletal System*. 2nd ed. Philadelphia: Lea & Febiger; 1989.

Level of Evidence:

Book

Summary:

This book provides a more in-depth discussion of biomechanics.

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